RAHUL PAHUJA

Staff Mobile Engineer | Android & iOS Architecture | Mobile Security & Cryptographic SDKs

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PROFESSIONAL SUMMARY

Staff Mobile Engineer with 11+ years of systems-level expertise across Android and iOS, specializing in hardware-backed mobile security, SDK architecture, and performance engineering. Architected the enterprise zero-trust storage layer (TeeSDK) and offline architecture foundation for **Idira by Palo Alto Networks (formerly CyberArk)**, unblocking **$240M in enterprise contract value** and eliminating authentication bypass vectors across 1M+ active enterprise seats. Proven track record driving STRIDE threat modeling, authoring Architecture Decision Records (ADRs), modeling complex systems via C4 architecture diagrams, benchmarking tail-latencies (P50/P95/P99), and standardizing client-side engineering practices across cross-functional organizations.

$240M Contract Value

Unlocked enterprise deal pipeline via hardware-backed offline storage architecture baseline.

1M+ Active Users

Secured via cryptographic biometric tampering defense and customized DexGuard hardening.

28% Cold-Start Latency Cut

Achieved via P95/P99 I/O profiling, DataStore multi-layer caching, and optimized startup paths.

TECHNICAL EXPERTISE

Mobile Security	Android Keystore, iOS Secure Enclave, Hardware-backed Biometric Auth (CryptoObject), STRIDE Threat Modeling, OWASP Mobile Top 10, SSDLC, DexGuard, ProGuard/R8, Zero-Trust Storage.
Platforms & Langs	Kotlin, Swift, Java, Objective-C, Jetpack Compose, SwiftUI, UIKit, Coroutines & Flow, Combine, ExoPlayer, AVFoundation, C++ (JNI).
Architecture	C4 Architecture Modeling, Clean Architecture, Modularization, Architecture Decision Records (ADRs), Preferences vs. Proto DataStore, Hilt/Dagger/Koin, gRPC, Protobuf, REST, GraphQL, Offline Sync.
Performance & Edge	P50/P95/P99 latency profiling, ANR & OOM elimination, multi-threading concurrency, Core ML, TensorFlow Lite, on-device ML/OCR inference, CI/CD automation (Fastlane, GitHub Actions).

PROFESSIONAL EXPERIENCE

Independent Engineering Lab & Advisory (Mobile1x) | Principal Systems Architect

Dec 2025 – Present | Indore / Remote

- **On-Device Privacy Products (Dual-Store Deployments):** Architected and published dual-platform native apps enforcing 100% on-device data sovereignty (600+ users): DeDup (Android Play Store) / (iOS App Store) and Notes — Archive & Reflect (Android) / (iOS) with on-device speech-to-text, Core ML OCR, and zero cloud telemetry.
- **E-Commerce Architecture & Scale (2.5K DAU):** Architected the full-stack mobile client and checkout infrastructure for The A2Z Collection integrating Razorpay payment gateways, automated carrier shipping APIs, resilient OTP phone authentication, and Cloudflare R2 object storage for zero-egress asset caching; sustained 2,500 Daily Active Users (DAU).
- **Enterprise Form Engine:** Architected *Smart Form*, a schema-driven dynamic JSON form engine for enterprise data capture, cutting engineering form-creation cycles by **50%**.

Idira by Palo Alto Networks (formerly CyberArk) | Staff Software Engineer - Mobile Architecture

Nov 2024 – Dec 2025 | Hyderabad, India

Live Apps: Idira Identity (iOS App Store) • Idira (Google Play Store)

- **$240M Commercial Contract Impact:** Architected TeeSDK, a zero-trust hardware-backed secure storage foundation across Android Keystore and iOS Secure Enclave; established the core architectural baseline for the enterprise offline feature, unblocking strict compliance requirements and **unlocking $240M in enterprise contract value** across 1M+ active enterprise seats.
- **C4 Modeling & ADR Standardization:** Mapped end-to-end mobile security topologies across Context, Container, Component, and Code levels using C4 architecture diagrams; authored foundational ADRs establishing clear design governance across distributed mobile, backend, and security stakeholders.
- **Zero-Day Biometric Defense:** Engineered hardware-enforced biometric authentication using Android Keystore crypto-

PROFESSIONAL EXPERIENCE (CONTINUED)

Conviva | **Senior Engineer - Android Sensors & Telemetry SDK** — **May 2022 – Sep 2023** | **Bengaluru, India**

- **Massive Real-Time Telemetry:** Engineered ExoPlayer telemetry sensor integration for real-time analytics platform calculating **12B+ metric computations per minute** across top global video streaming platforms.
- **45% Playback Incident Drop:** Resolved long-standing CSAI/SSAI ad-tracking discrepancies and race conditions, resulting in a **45% reduction** in client-reported playback incidents.
- **Concurrency Redesign:** Re-architected multithreading concurrency models for Tier-1 enterprise clients (ShareChat), eliminating critical out-of-memory (OOM) conditions and race-condition crashes at scale.

Intellicus Technologies (Kyvos) | **Senior Software Engineer** — **Dec 2019 – Apr 2022** | **Bengaluru, India**

Live App: Intellicus Mobile (Google Play Store)

- **Stability & Speed:** Drove Java-to-Kotlin modernization and Firebase Crashlytics observability, cutting application crash rates by **32%** and improving OLAP analytics dashboard load times by **40%**.
- **SDK Development:** Built the core Intellicus Mobile SDK for embedded OLAP business intelligence reporting on Android; instituted JUnit unit testing and architectural modularization standards across legacy codebases.

Wipro Limited | **Software Engineer to Senior Project Engineer** — **Aug 2014 – Dec 2019** | **Bengaluru, India**

- **Connected Vehicles (Mahindra First Choice Wheels — AutoInsights):** Delivered an industry-first IoT connected car telematics platform powering *Word of Car*, real-time *Health of Car* diagnostics, driving analytics, and pioneering **Live Location Sharing** ahead of mainstream adoption by WhatsApp, Ola, and Uber across Android and iOS clients.
- **Enterprise iOS Delivery (Chevron & ELCITA):** Architected and delivered native iOS applications including the industrial Chevron Lubricant Track & Trace app on the Apple App Store for enterprise supply chain visibility, as well as the civic ELCITA Citizen iOS app for Electronic City Bengaluru; automated SSIS data migration interfaces, cutting enterprise migration times by **95%**.

SELECTED ARCHITECTURAL & SECURITY HIGHLIGHTS

- **C4 System Design & Architectural Governance:** Utilized C4 model diagrams (Context, Container, Component, Code) to document multi-tier mobile client integrations, bridging technical implementation detail with executive-level system visualization.
- **TeeSDK Cryptographic Foundation:** Hardware-backed key attestation, key lifecycle management, zero-trust credential isolation, and secure offline caching across Android Keystore and iOS Secure Enclave.
- **Data Persistence ADRs:** Multi-tiered benchmarks across cold starts, thread contention, and serialization trade-offs, establishing definitive engineering standards for local persistence.
- **Binary Defense in Depth:** Threat-model-driven obfuscation strategy using DexGuard for control-flow virtualization, string decryption, and root detection without degrading cold-start SLA budgets.

TECHNICAL LEADERSHIP, TEACHING & RECOGNITION

- **Google-Affiliated Engineering Bootcamp (LTIMindtree):** Designed curriculum and mentored 41 Android engineers from core systems to production architectures, achieving 100% placement.
- **Technical Writing & Architecture Research:** Author of deep-dive analyses on mobile security, on-device edge ML, and coroutine architectures published at *therahulpahuja.com*.
- **API World Award:** Recognized at API World for outstanding Open Banking architecture delivered on mobile clients.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani — *M.Tech, Software Engineering / Computer Science* — **2015 – 2019**

School of Computer Science & I.T, Indore — *BCA, Computer Science & Applications* — **2011 – 2014**